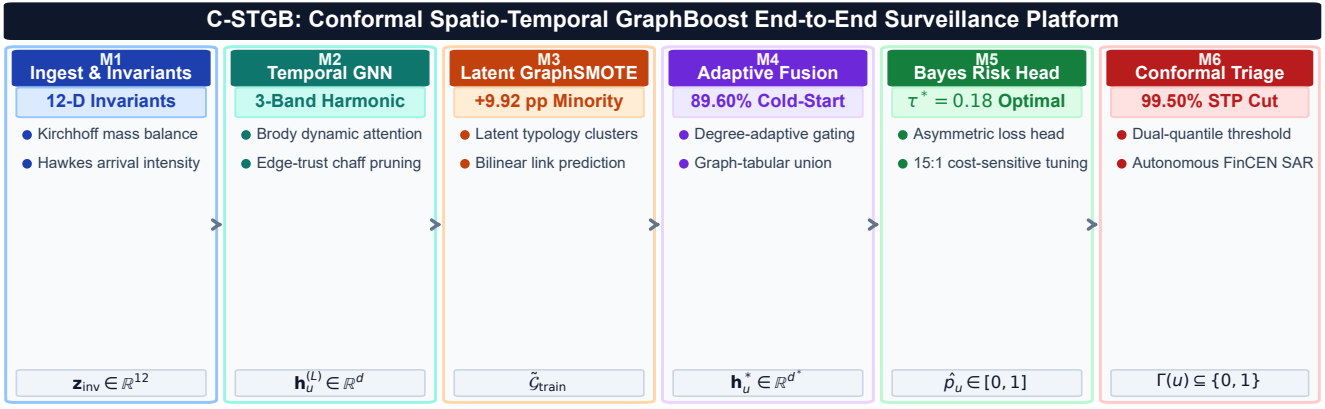

C-STGB: Conformal Spatio-Temporal GraphBoost End-to-End Surveillance Platform
M1
Ingest & Invariants
12-D Invariants
Kirchhoff mass balance
Hawkes arrival intensity
$\mathbf{z}_{\text{inv}} \in \mathbb{R}^{12}$
M2
Temporal GNN
3-Band Harmonic
Brody dynamic attention
Edge-trust chaff pruning
$\mathbf{h}_u^{(L)} \in \mathbb{R}^d$
M3
Latent GraphSMOTE
+9.92 pp Minority
Latent typology clusters
Bilinear link prediction
$\tilde{\mathcal{G}}_{\text{train}}$
M4
Adaptive Fusion
89.60% Cold-Start
Degree-adaptive gating
Graph-tabular union
$\mathbf{h}_u^* \in \mathbb{R}^{d^*}$
M5
Bayes Risk Head
$\tau^* = 0.18$ Optimal
Asymmetric loss head
15:1 cost-sensitive tuning
$\hat{p}_u \in [0, 1]$
M6
Conformal Triage
99.50% STP Cut
Dual-quantile threshold
Autonomous FinCEN SAR
$\Gamma(u) \subseteq \{0, 1\}$